

● EASY

● ADVANCED MATH

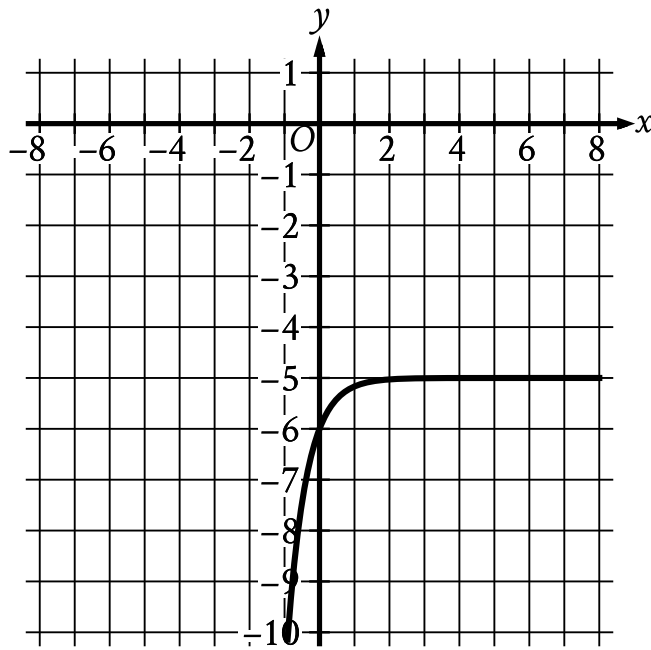
Advanced Math

05

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29



- Moving from left to right:
 - The curve passes from quadrant 3 to quadrant 4.
 - In quadrant 3, the curve trends up sharply to point (0 comma negative 6).
 - In quadrant 4, the curve trends up gradually.
- As x increases, the curve approaches the line y equals negative 5.
- The curve passes through the following points:
 - (0 comma negative 6)
 - (1 comma negative 5 and one sixth)

What is the y-intercept of the graph shown?

- A. 0, -6
- B. -6, 0
- C. 0, 0

D. -5, - 5

Q2

ADVANCED MATH

$$P = \frac{W}{t}$$

The power P produced by a machine is represented by the equation above, where W is the work performed during an amount of time t . Which of the following correctly expresses W in terms of P and t ?

A. $W = Pt$

B. $W = \frac{P}{t}$

C. $W = \frac{t}{P}$

D. $W = P + t$

Q3

ADVANCED MATH

Which expression is equivalent to $2x^2 + x - 9 + x^2 + 6x + 1$?

- A. $2x^2 + 7x + 10$
- B. $2x^2 + 6x - 8$
- C. $3x^2 + 7x - 10$
- D. $3x^2 + 7x - 8$

Q4

ADVANCED MATH

$$Pt = 1,800(1.02)^t$$

The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. What is the best interpretation of $P_0 = 1,800$ in this context?

- A. The estimated number of marine mammals in the area was 102 when the study began.
- B. The estimated number of marine mammals in the area was 1,800 when the study began.
- C. The estimated number of marine mammals in the area increased by 102 each year during the study.
- D. The estimated number of marine mammals in the area increased by 1,800 each year during the study.

Q5

ADVANCED MATH

$$x + 5 = 14$$

$$y = 4x^2 + 4$$

At what point x, y do the graphs of the equations in the given system intersect?

- A. 9, 324
- B. 9, 328
- C. 14, 4
- D. 14, 788

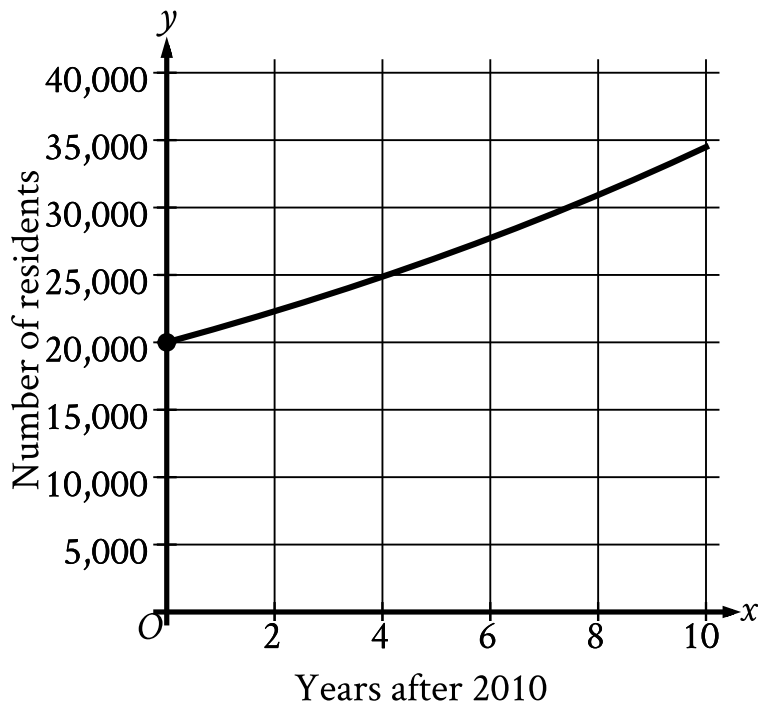
Q6

ADVANCED MATH

$$(2x^3 + 3x)(x^3 - 2x)$$

Which of the following is equivalent to the expression above?

- A. $x^3 + 5x$
- B. $3x^3 + x$
- C. $2x^6 - x^4 - 6x^2$
- D. $3x^6 - x^4 - 6x^2$



- The curve is in quadrant 1.
- The curve trends gradually up from left to right.
- The curve begins at the point (0 comma 20,000).
- The curve passes through the following points:
 - (0 comma 20,000)
 - approximately (5 comma 26,263)
 - approximately (10 comma 34,488)

The graph shown models the number of residents of a certain city x years after 2010. How many residents does this model estimate the city had in 2010?

- A. 0
- B. 2,000
- C. 20,000

D. 25,000

Q8

ADVANCED MATH

Which of the following is a solution to the equation $2x^2 - 4 = x^2$?

A. 1

B. 2

C. 3

D. 4

Q9

ADVANCED MATH

Which expression is equivalent to $9x + 6x + 2y + 3y$?

A. $3x + 5y$

B. $6x + 8y$

C. $12x + 8y$

D. $15x + 5y$

The function h is defined by $hx = \frac{8}{5x+6}$. What is the value of $h2$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.